

Jeffrey Shi

jshi430.github.io | github.com/js430

Active TS/SCI with Polygraph

Email: jeffreyshi430@gmail.com

Tysons Corner, VA

Cyber threat analyst and data scientist with 5 years supporting U.S. Intelligence Community cyber missions. Combines network and SIGINT analysis, MITRE ATT&CK-based TTP analysis, and Python/SQL at scale with hands-on experience building LLM-driven workflows to accelerate investigations and reporting.

EXPERIENCE

GRVTY

Data Scientist / Network Analyst

Tysons Corner, VA

April 2026 – Present

- **Cyber Mission Support:** Analyze collected network data and SIGINT with GOTS analytic and targeting tools to identify adversary cyber activity, mapping observed TTPs to MITRE ATT&CK in support of a USIC agency.
- **Network Forensics:** Perform packet and flow analysis across large network datasets to surface anomalous and malicious behavior and inform finished intelligence reporting.
- **AI/LLM Tooling:** Built LLM-based workflows to triage and process large network-related datasets, cutting time from collection triage to finished report by 50%+ and enabling the team to produce more reporting over more data in the same period.
- **Mission Applications:** Design and stand up custom one-off applications that support day-to-day operations for the IC customer.

ManTech

ETL/Data Engineer

Herndon, VA

Nov 2024 – April 2026

- **Security Data Pipelines:** Developed modular Python ETL pipelines that parse and normalize output from 5+ industry-standard security scanners (Titania Nipper, Rapid7, Nmap, and others) for ingest into an IC customer's cyber assessment platform; built the Titania Nipper module end to end.
- **Endpoint Collection:** Researched and built custom Velociraptor artifacts to collect host configuration data used to assess the security posture and compliance of assets and devices.
- **PostgreSQL:** Wrote maintenance scripts to eliminate database inefficiencies and resolve data issues affecting customer-facing UI components.

General Dynamics Information Technology (GDIT)

ETL/Data Engineer

McLean, VA

Oct 2023 – Nov 2024

- **Streaming ETL:** Built and maintained asynchronous Python (asyncio) pipelines consuming Kafka topics and loading ~2.7M records/day (1B+ per year) into Oracle and PostgreSQL for a live IC mission data feed.
- **Platform Migration:** Helped migrate legacy Pentaho ETL jobs to a containerized Python framework on Kubernetes; refactored a FastAPI/Neo4j service to visualize end-to-end data flow.

Cloud Developer/Engineer

Sept 2022 – Oct 2023

- **Open-Source Collection:** Automated recurring web data collection with Python and Selenium, transforming results with pandas and loading them into an AWS RDS Oracle database for IC analysts and data scientists.
- **Data Integrity:** Built a monthly Python/SQL (cx_Oracle) validation job that flagged and corrected bad values across database tables.

Accenture LLP

Frontend Developer / Product Delivery Lead

Boston, MA / Remote

July 2021 – Sept 2022

EDUCATION

University of Virginia

B.A. Computer Science, B.A. Statistics; Echols Scholar

Charlottesville, VA

Aug 2018 – May 2021

SKILLS & CERTIFICATIONS

Threat Intelligence: MITRE ATT&CK, TTP analysis, network/SIGINT analysis, packet & flow analysis, GOTS analytic and targeting tools, intelligence reporting

AI/LLM: LLM-driven triage and analysis workflows over large datasets

Languages & Data: Python, SQL (PostgreSQL, Oracle), Bash, pandas, Kafka, asyncio, FastAPI, Neo4j, Selenium

Security Tools: Velociraptor, Rapid7, Nmap, Titania Nipper

Cloud & DevOps: AWS (RDS, EC2, S3), Google Cloud (BigQuery, GKE), Kubernetes, Docker, GitLab, Jenkins

Certifications & Other: AWS Certified Cloud Practitioner; Chinese (fluent)